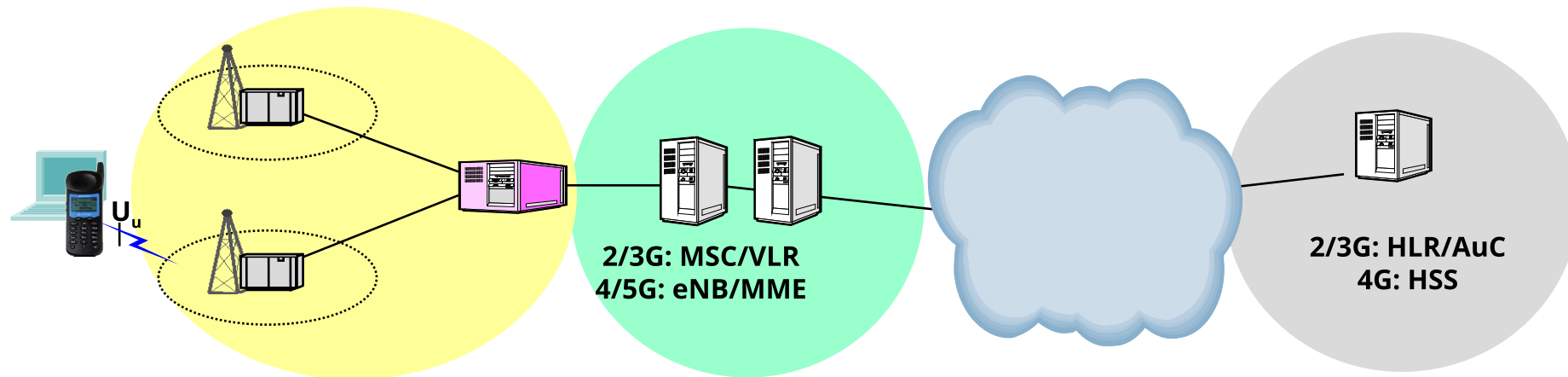


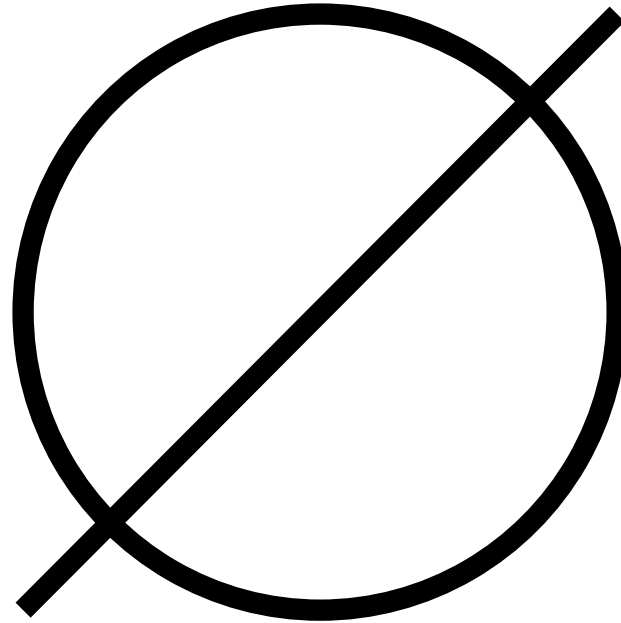
Security in Wireless Cellular systems

(focus limited to authentication, at this stage)

Architecture of wireless cellular systems (abstracted)



Security in 1G



Security in 2G

Security? No problem!
Let me do it!!



Can you
Protect us?



COMP-128 Security by obscurity

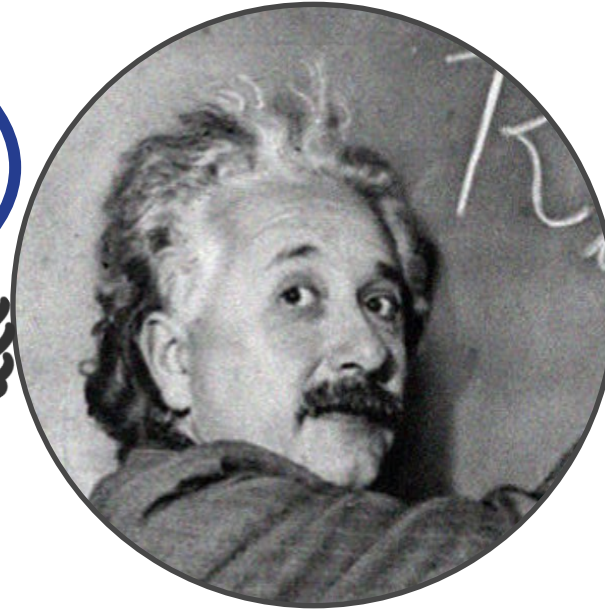
No mutual auth (Rogue BS)

No core network security



Security in 3G

OK, let's be serious now...



Pro to the rescue!!



(fairly) good ciphers - public scrutiny!

Encryption AND (in part) Integrity

Mutual authentication

Core network security

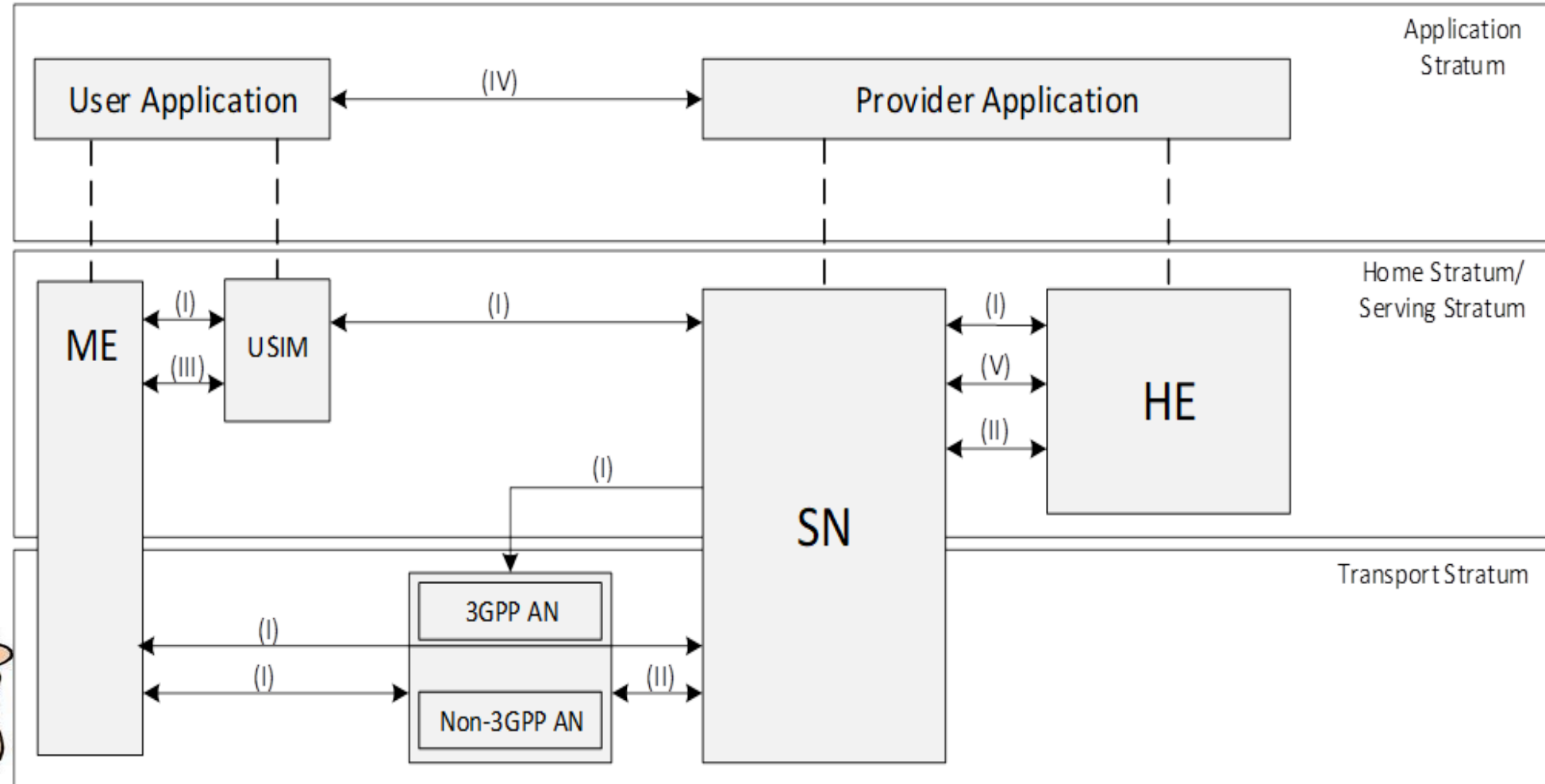
Multiple keys

Security in 4G

Time for a
Security
architecture!



Systematic
approach



Security in 4G and 5G

Actually... this is already the 5G security architecture... to save one slide (couple of differences over 4G)!

(VI) Visibility and Configuration of Security

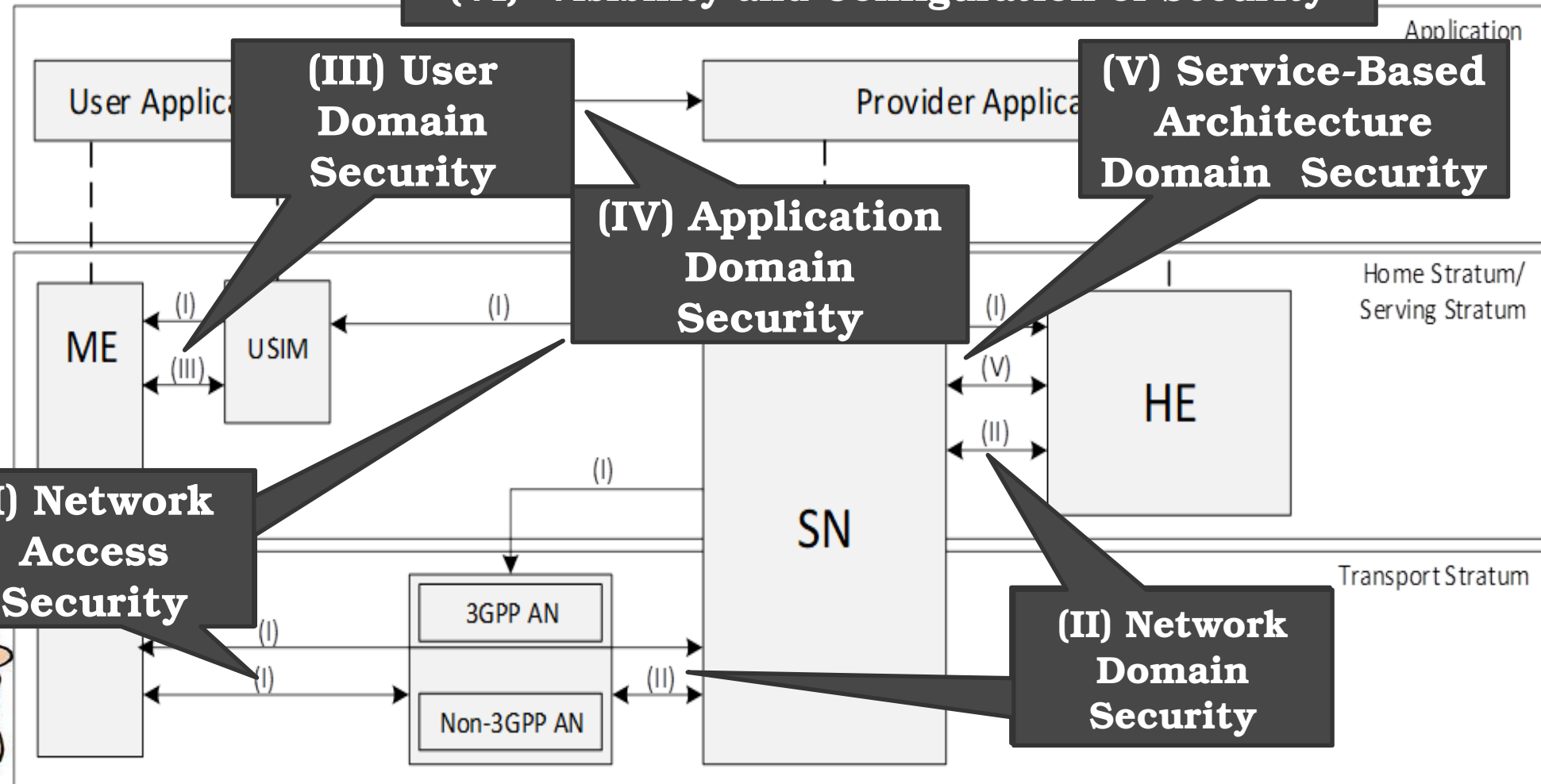
(III) User Domain Security

(IV) Application Domain Security

(V) Service-Based Architecture Domain Security

(I) Network Access Security

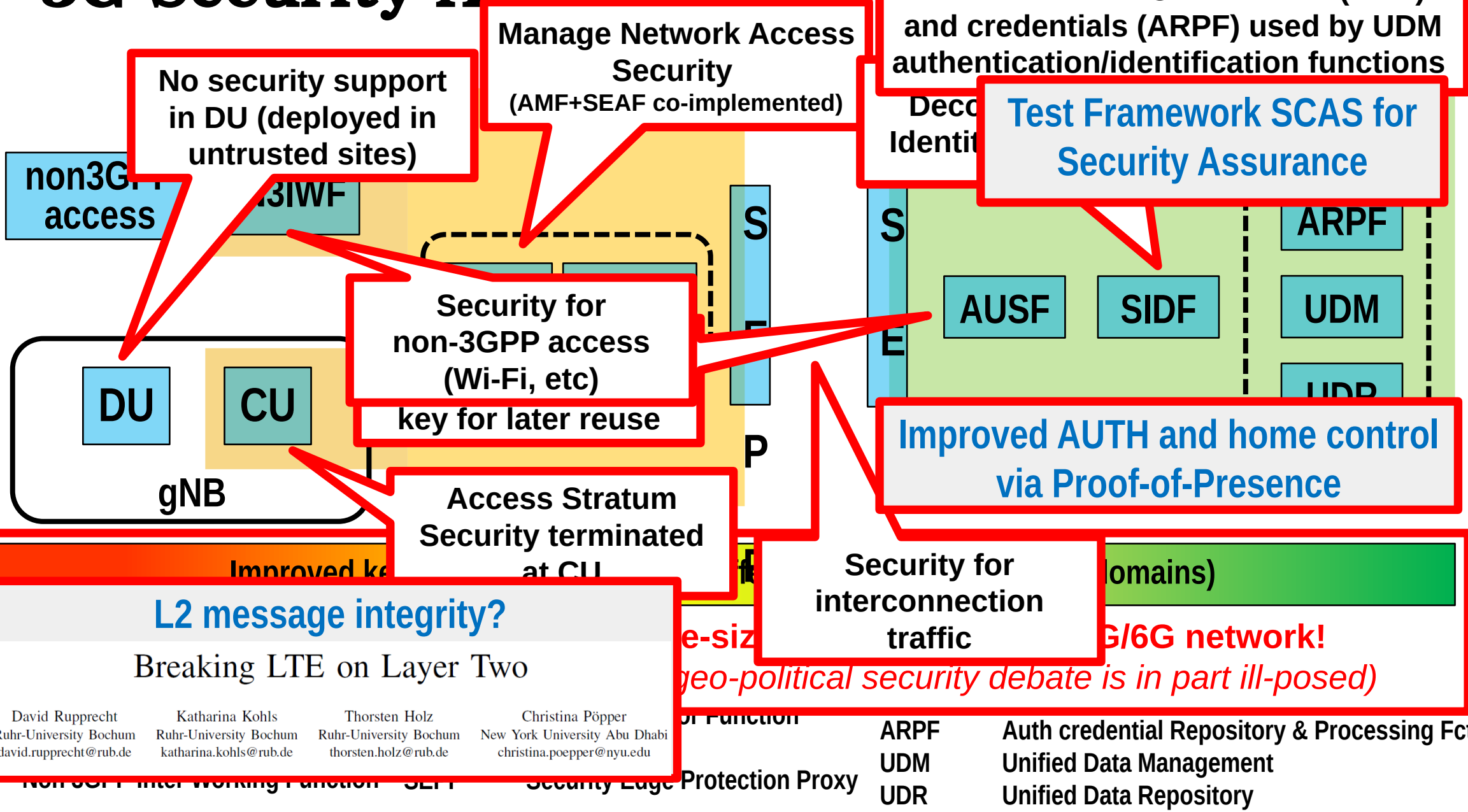
(II) Network Domain Security



Time for a Security architecture!

Systematic approach

5G Security Architecture



L2 message integrity?

Breaking LTE on Layer Two

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But sys

SECURITY IN 5G SPECIFICATIONS

Controls in 3GPP Security Specifications (5G SA)

FEBRUARY 2021

Complex

**5G/6G
systems' security:**
A small part of a
bigger picture!!

g security

Firmware

**5G/6G
security**



SECURITY IN 5G SPECIFICATIONS
February 2021

**No 5G security !!
SW/HW implementation
is insecure!**

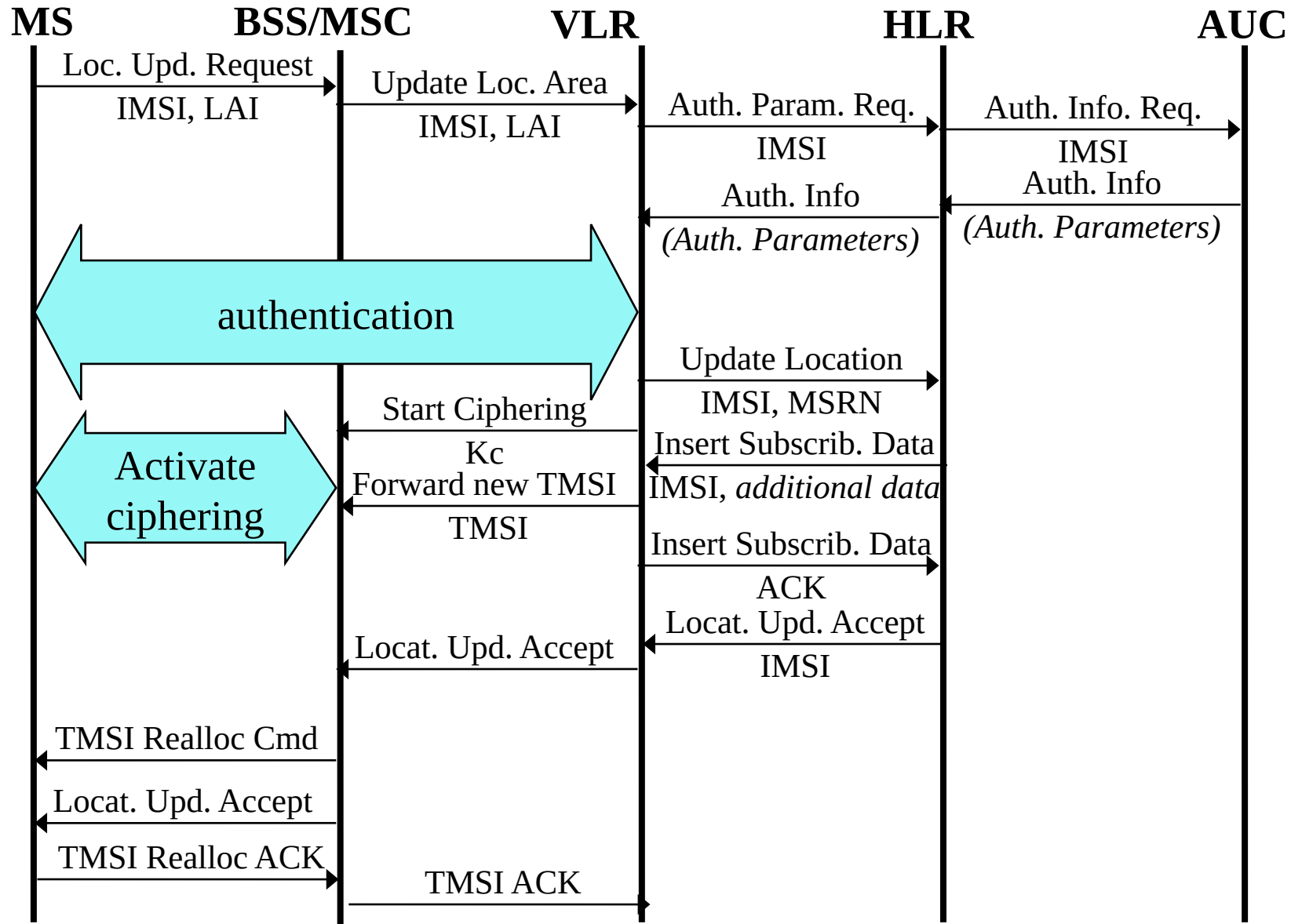
**Critical infrastructure = high tier threats
«They» might design vulnerabilities!**

**SW networks, third party suppliers:
Complexity, increased threat surface!**

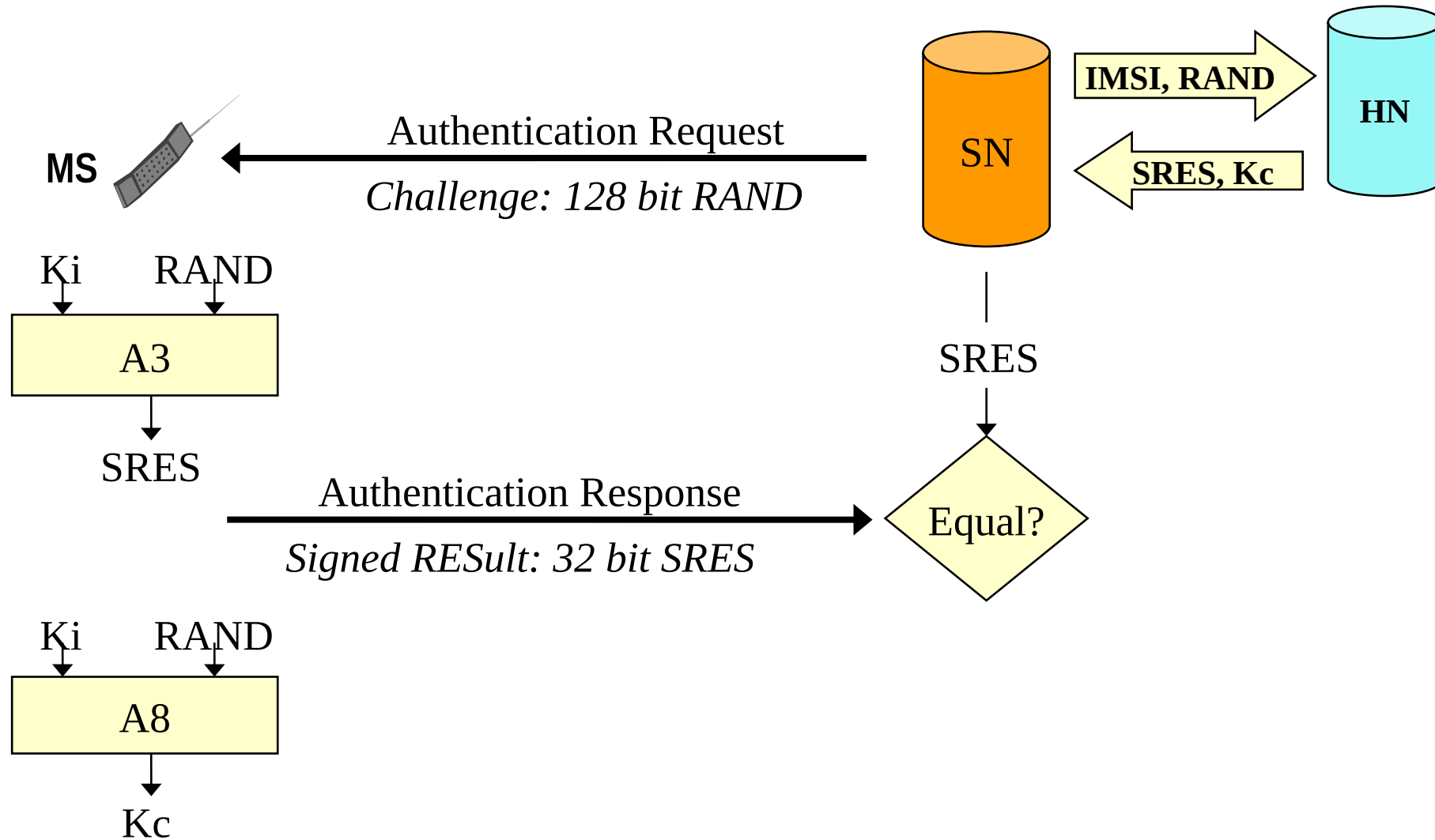
**Too many «optional/should»
In the standard!**
Very risky! (you might believe you are secured, while...)

Challenge-Response authentication in Wireless Cellular systems

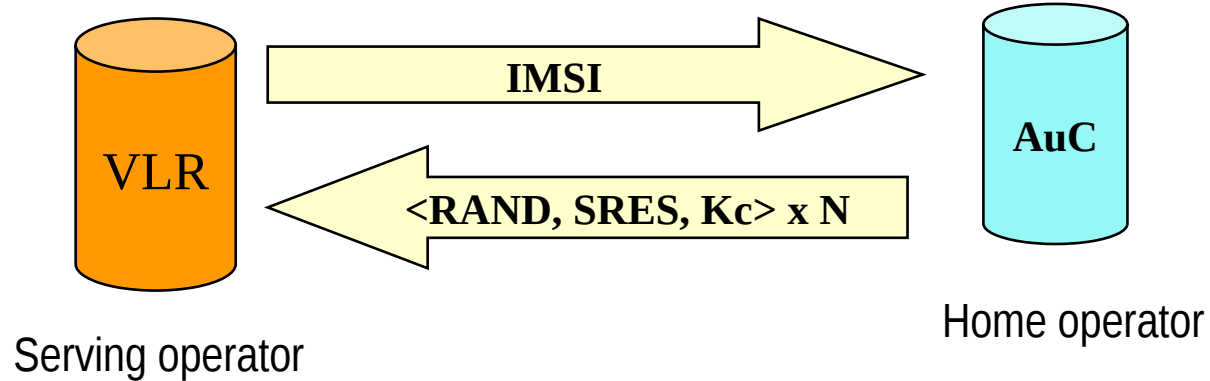
Authentication: when (e.g. 2/3G)



Authentication in 2G (unilateral)

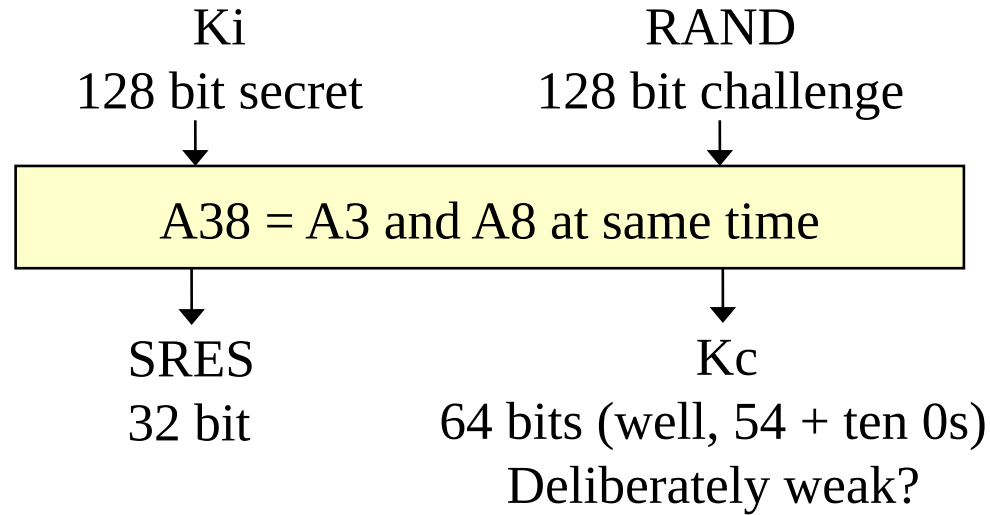


Triplets (Authentication Vector)



- **Idea: once in a VLR area, authentication will need to be performed MANY times**
- **Hence deliver N triplets, to be used for N distinct authentications**
- **IMPORTANT: VLR does NOT need to know authentication algo used (A3, A8)**
 - ⇒ Triplet contains computed result by AuC
 - ⇒ A3, A8 run inside the SIM (given by operator)

Authentication: details



→ Challenge response with:

⇒ Challenge

→ RAND

⇒ Secret

→ Ki

⇒ Hash

→ A3 algorithm

On the A3/A8 algorithms

→ Security by obscurity

- ⇒ A3 algorithm CAN BE operator-specific!
- ⇒ But most vendors originally used algo today called COMP128
- ⇒ Non disclosed but...
 - Reverse engineered? Leaked out?

→ COMP128 broken in 1998

- <http://www.isaac.cs.berkeley.edu/isaac/gsm-faq.html>
- ⇒ Chosen challenge attack [1998, Briceno, Goldberg, Wagner]
 - Ki disclosed through about 150.000 queries with suitably selected challenges → approx 8h in 1998
 - Having Ki means cloning the card!
- ⇒ Better attack [2002, Rao, Rohatgi, Scherzer, Tinguely]
 - Less than 1 minute!

→ Lesson learned:

- ⇒ Security by obscurity does NOT work!! Leave hash design to crypto experts

Over the air attach

→ **No mutual authentication!**

→ **Rogue BTS may easily perform the attach**

⇒ run it for sufficient time

3G/4G/5G authentication: AKA Authentication and Key Agreement

(STRONGLY Simplified for our purposes; 3G terminology)

Major differences with 2G

→ **Mutual authentication!**

⇒ Optimized...

⇒ Guaranteed freshness for auth parameters

→ **More comprehensive security**

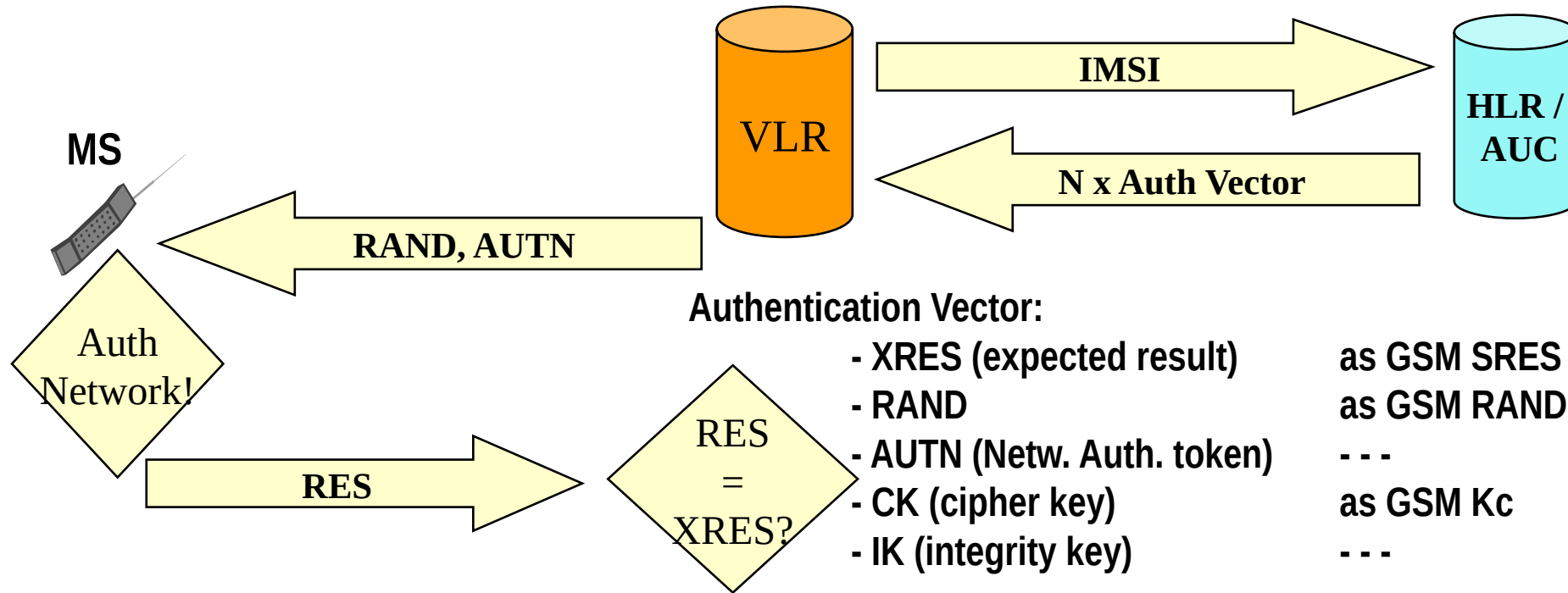
⇒ More keys and several extra details

→ (we will not focus on this)

→ **Algorithms FIRST scrutinized by research community, THEN selected**

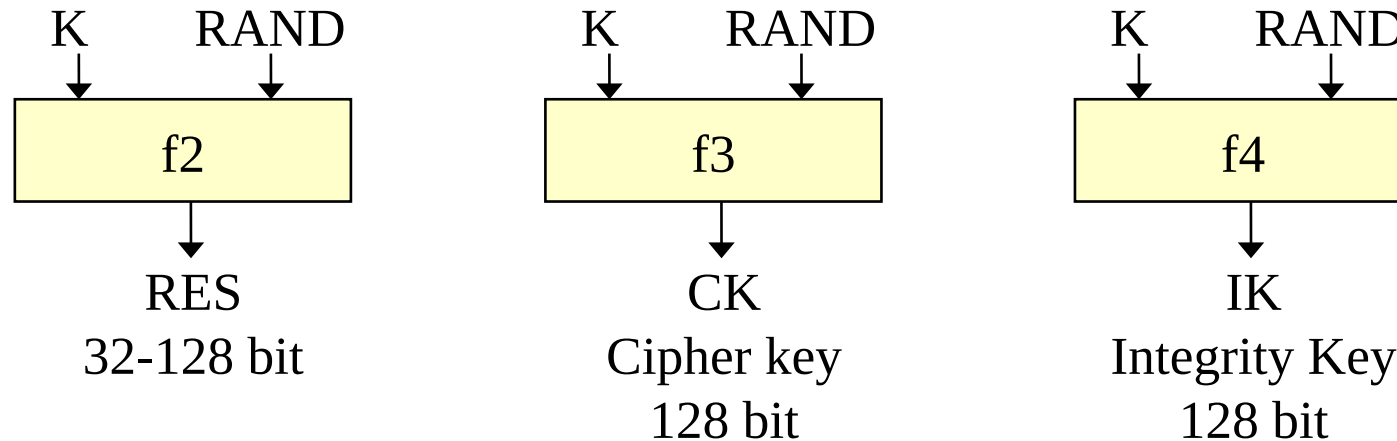
⇒ The opposite of security by obscurity 😊

Authentication Vector



MS authentication

➔ Usual challenge response, but different (public) algorithm



Network Authentication

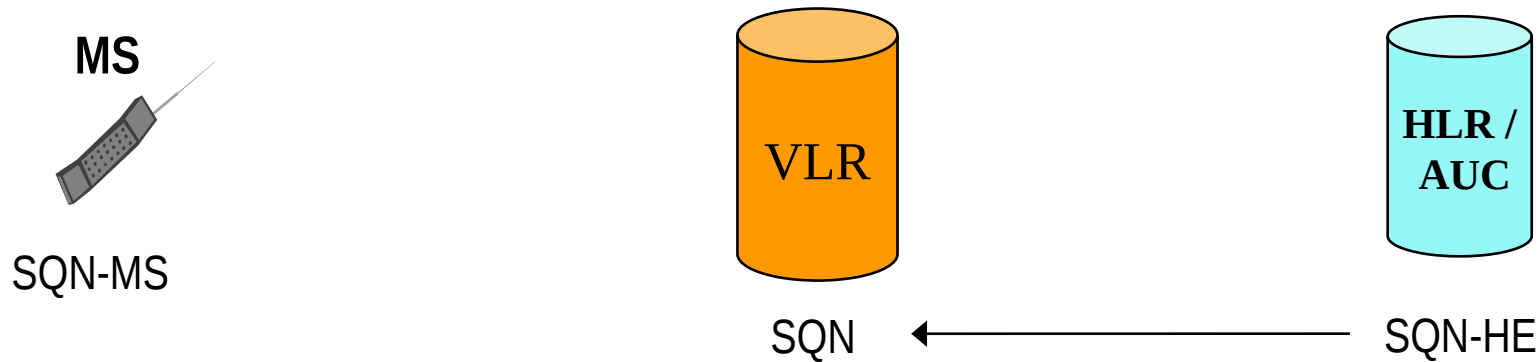
→ **MS should send a nonce...**

⇒ 1 extra message

→ **Bright idea: use Sequence number as “implicit” nonce!**

⇒ Issue: MS and AuC must be (approx) synchronized

⇒ And robust procedures for resync must be specified



SEQ as nonce: idea

Current SQN-MS
stored

Check
 $SQN = SQN-MS + 1$
(or in appropriate
Tolerance range to
cope with lost msg)

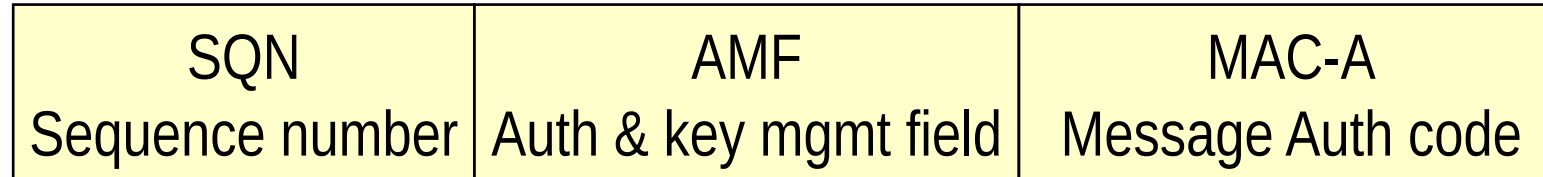
SQN
(included in AUTN)
(given by AuC)



Use SQN as
“implicit” challenge!!

Once auth OK, update local SQN

Network authentication: AUTN format

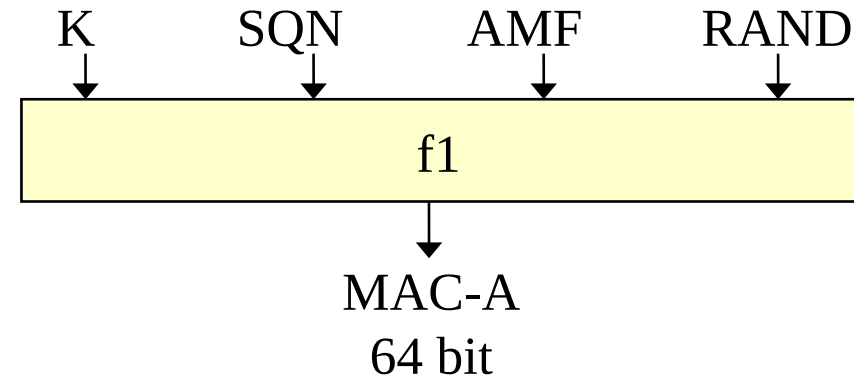


48 bit

16 bit: carries info on
which algo or key to
use if choice available,
sync window, etc
(signalling info)

64 bit: allows MS to
verify authenticity of
Network!

**MS: has all info needed to check
that MAC-A transmitted by network
is the same of MAC-A locally computed!
SQL guarantees defense against replay**



Minor detail: protecting SQN!

→ A privacy problem:

⇒ By looking at SQN (stepwise increasing), eavesdropper may discriminate and track user!

→ Solution: mask SQN with Anonymity Key

SQN xor AK Sequence number	AMF Auth & key mgmt field	MAC-A Message Auth code
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